

Gaurav Abhay Desai

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Summary

Results-driven software engineer with expertise in software engineering, machine learning, AI, and application development, specializing in backend systems and data-driven solutions. Skilled in optimizing infrastructures, enhancing performance, and driving system innovation and automation.

Skills

- **Languages:** Python, Java, C#, golang, JavaScript, C/C++, Shell Scripting, Solidity, GraphQL, Bash, HTML
- **Frameworks/Libraries:** SpringBoot, .NET, Node.js, React.js, Express.js, TailwindCSS, NestJS, Next.js
- **Databases:** MySQL, Postgres, Redis, MongoDB, MS-SQL, NoSQL, SQL Server
- **DevOps/Tools/Other:** Apache Kafka, Elasticsearch, Docker, Kubernetes, Prometheus, Ansible, Jenkins, Git, TFS, Generative AI, Deep Learning, Machine Learning, Artificial Intelligence, Linux Operating System, Rest APIs, Agile Methodology, Object Oriented Programming, JSON, SDLC
- **Cloud Technology:** AWS, Azure, Google Cloud, Firebase
- **Certification:** Google Cloud Computing Foundations Certificate

Professional Experience

MOFSL | *Software Developer*

Nov 2022 - Jun 2024

- Engineered real-time market data pipelines using Kafka, Redis, and Solace to process 5M+ messages daily with sub-millisecond latency, boosting decision-making speed for traders.
- Developed backend APIs in C#, Java and Go for order execution and risk checks, ensuring secure, scalable interactions that supported 100K+ daily transactions without performance bottlenecks.
- Automated recurring monitoring and cleanup tasks with Python scripting integrated into scheduled jobs, reducing manual interventions and cutting maintenance overhead by 25%. Enhanced fault tolerance of the trading system by introducing retry and failover mechanisms, improving system uptime and reducing downtime incidents by 20%.
- Collaborated cross-functionally with trading desk and compliance teams to capture requirements, translate them into technical solutions, and deliver features ahead of trading deadlines.

CHCS Services Inc. | *Process Associate*

Sep 2021 - Nov 2021

- Built automated database validation scripts in Python and SQL, eliminating manual error checks and reducing processing effort by 40%.
- Conducted routine checks and optimizations using Prometheus and Elasticsearch enhancing system performance and reducing downtime

Jaldoot Enterprises Ltd. | *Technical Partner (Software & Operations)*

Oct 2013 - Jul 2021

- Increased revenue by 25% YoY by launching a wealth sub-broking vertical, setting up KYC/advisory workflows, and converting ~200 existing customers into AUM clients.
- Reduced month-end close from 5 days to 1 day by digitizing all financial and inventory records, adding maker-checker controls, and standardizing SKUs.
- Built a lightweight personnel-tracking app in C# WinForms to log shifts, attendance, and field activity; integrated payout rules to automate rostering, reducing manual effort and improving payroll accuracy by 40%.
- Designed an HR tracking tool for employee attendance and payroll, reducing reporting errors by 30% and cutting manual bookkeeping efforts. Introduced weekly KPI dashboards (sales, AUM pipeline, stock-outs, collections aging) and used insights to clear slow-moving items, boosting working capital efficiency.
- Formalized customer retention workflows with CRM-style follow-ups, improving repeat orders and referrals while reducing escalations. Established SOPs for receiving, storage, and dispatch with approval thresholds, preventing errors and creating audit-ready traceability.

Education

California State University - Fullerton

Aug 2024 - May 2026

Master's, Computer Science (GPA: 3.93)

- **Coursework:** Algorithms, Machine Learning, Software Engineering, Advance Blockchain

Projects

Transforming Real Estate with AI | [Live Portal link](#)

- Built an AI-powered full-stack MERN real estate platform with JWT authentication, responsive listings, and Python FastAPI ML microservices using Random Forest and KNN for fair-price prediction, comparable-property search, and personalized, risk-aware investment recommendations.

BlogSpace | [Live Blog link](#)

- Built BlogSpace, an AI based MERN-stack blogging platform using React, Node.js, Express, and MongoDB with REST APIs and JWT authentication; developed an author dashboard with a React Quill editor and autosave drafts, implemented publish and scheduling workflows with server-side handling, integrated Cloudinary for image uploads, and delivered a responsive UI with search, filters, tags, reactions, comments, bookmarks, view tracking, and light/dark theme persistence.

AetherVault | [Github link](#)

- Developed a decentralized NFT auction and marketplace on Ethereum with minting, fixed-price sales, real-time auctions, and analytics dashboards for auction insights. Tech: Solidity, OpenZeppelin ERC-721, Truffle, React.js, ethers.js, IPFS (Pinata)

E-Adoption Portal | [Github link](#)

- Engineered a secure online adoption system with Java Spring Boot, SQL and React, featuring role-based access for users and administrators, streamlined adoption processes, and real-time orphanage management.